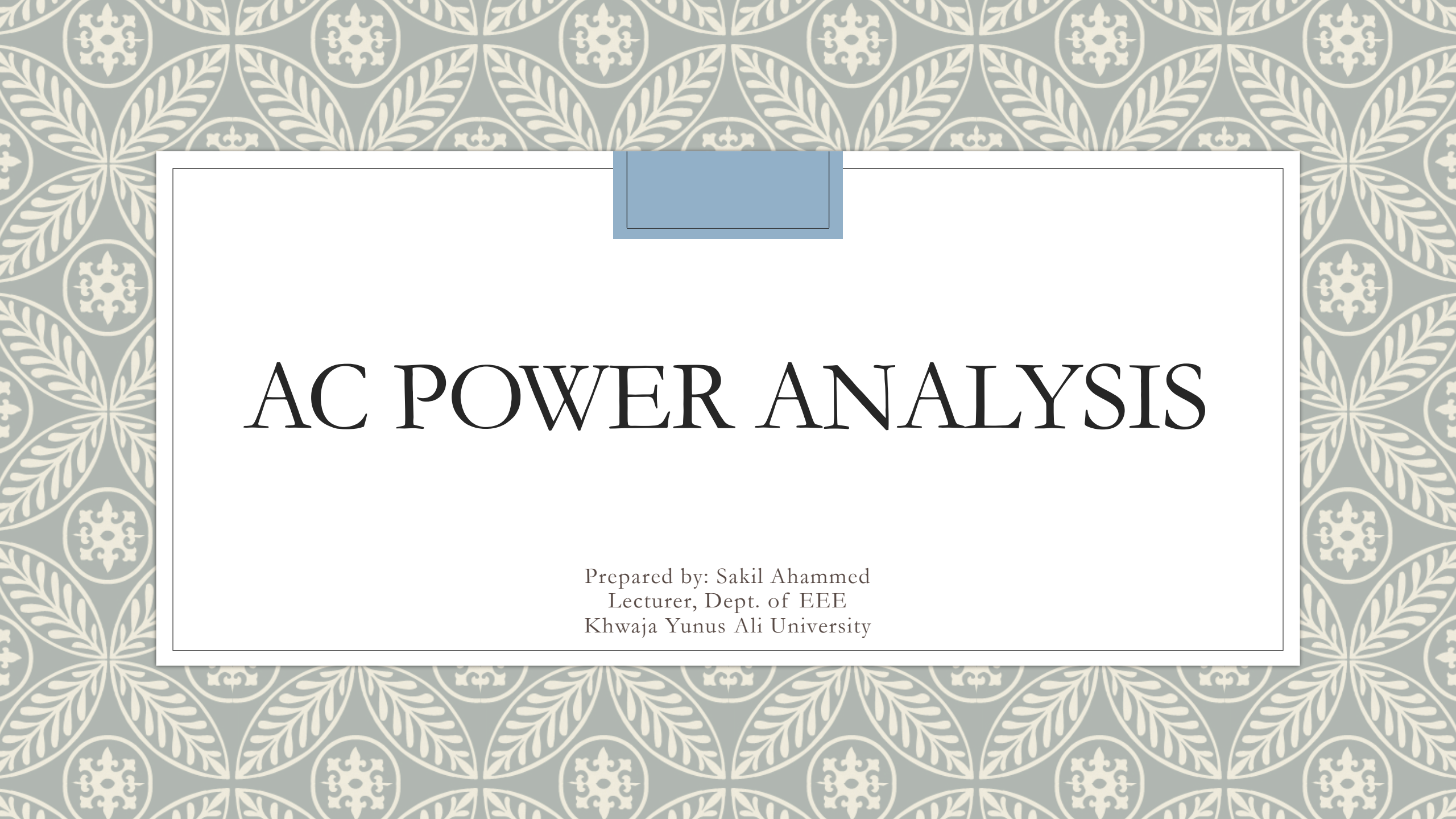


AC POWER ANALYSIS

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Introduction

- Power is the most important quantity in electric utilities, electronic, and communication systems, because such systems involve transmission of power from one point to another point.
- Every industrial and household electrical device (i.e. every fan, motor lamp, pressing iron, TV, personal computer) has a power rating that indicates how much power the equipment requires. Exceeding the power rating can do permanent damage to an appliance.
- The most common form of electric power is 50 Hz (in Bangladesh) or 60 Hz (in India) ac power.
- Ac is used over dc because ac allowed high voltage power transmission from the power generating plant to the consumer.

Instantaneous Power

- Instantaneous power $p(t)$ absorbed by an element is the product of the instantaneous voltage $v(t)$ across the element and the instantaneous current $i(t)$ through it.

$$p(t) = v(t) i(t)$$

- “The instantaneous power (in watts) is the power at any instant of time.”**
- It is the rate at which element absorbs energy. Instantaneous quantities are denoted by lower cases.

Let,

$$v(t) = V_m \cos(\omega t + \theta_v)$$

and,

$$i(t) = I_m \cos(\omega t + \theta_i)$$

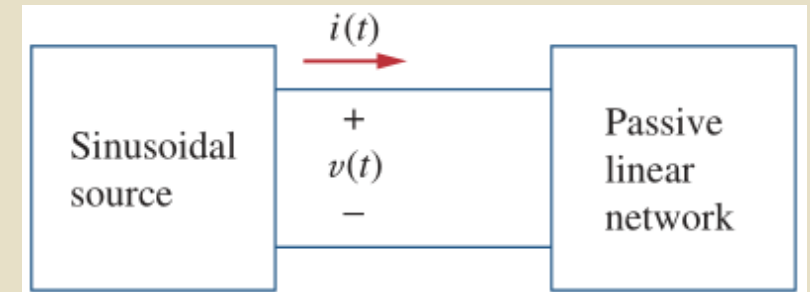
Where, V_m and I_m are peak values (or amplitudes) and θ_v and θ_i are phase angles of the voltage and current, respectively.

$$p(t) = v(t) i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) \quad \left(\text{As, } \cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]\right)$$

The instantaneous power has two parts, the first part is constant or time independent. Its value depends on the phase difference between the voltage and the current.

The second part is a sinusoidal function whose frequency is 2ω , which is twice the angular frequency of the voltage or current.



Average Power

- The instantaneous power changes with time and is therefore difficult to measure. As wattmeter measure average power, it is convenient to measure.
- **“The average power (in watts) is the average of the instantaneous power over one period.”**
- The average power is given by,

$$P = \frac{1}{T} \int_0^T p(t) dt$$

Substituting the value of instantaneous power $p(t)$ from the previous slide, we get

$$P = \frac{1}{T} \int_0^T \left[\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) \right] dt$$
$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt$$

The second integrand is sinusoids, and we know the average of sinusoids is zero. Thus we can neglect the second term. But the first integrand is constant. Average of a constant is constant itself.

So,

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Average Power contd...

◦ In phasor form

$$\begin{aligned}\frac{1}{2}VI^* &= \frac{1}{2}V_m I_m \angle(\theta_v - \theta_i) \\ &= \frac{1}{2}V_m I_m [\cos(\theta_v - \theta_i) + j \sin(\theta_v - \theta_i)]\end{aligned}$$

Thus,

$$P = \frac{1}{2} \operatorname{Re}[VI^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

If, $\theta_v = \theta_i$,

$$P = \frac{1}{2} V_m I_m = \frac{1}{2} I_m^2 R = \frac{1}{2} |I|^2 R \quad [\text{where, } |I|^2 = I \times I^*] \quad (\text{For resistive circuit})$$

If, $\theta_v - \theta_i = 90^\circ$,

$$P = \frac{1}{2} V_m I_m \cos(90^\circ) = 0 \quad (\text{For purely reactive circuit, power absorption}=0)$$

Conclusion: A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

Problems

Given that

$$v(t) = 120 \cos(377t + 45^\circ) \text{ V} \quad \text{and} \quad i(t) = 10 \cos(377t - 10^\circ) \text{ A}$$

find the instantaneous power and the average power absorbed

Example 11.1

Calculate the instantaneous power and average power absorbed
if

$$v(t) = 330 \cos(10t + 20^\circ) \text{ V} \quad \text{and} \quad i(t) = 33 \sin(10t + 60^\circ) \text{ A}$$

Practice Problem 11.1

Calculate the average power absorbed by an impedance $\mathbf{Z} = 30 - j70 \, \Omega$
when a voltage $\mathbf{V} = 120 \angle 0^\circ$ is applied across it.

Example 11.2

Practice Problem 11.2

A current $\mathbf{I} = 33 \angle 30^\circ \text{ A}$ flows through an impedance $\mathbf{Z} = 40 \angle -22^\circ \, \Omega$.
Find the average power delivered to the impedance.

Example 11.3

For the circuit shown in Fig. 11.3, find the average power supplied by
the source and the average power absorbed by the resistor.

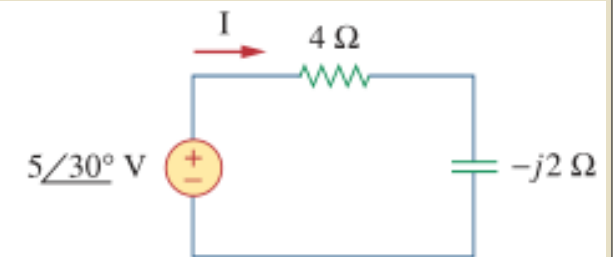


Figure 11.3

Effective and RMS value

“The effective value of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.”

Equations: $P = \frac{1}{T} \int_0^T i^2 R dt = I_{eff}^2 R$ (Where $I_{eff} = I_{rms}$, $V_{eff} = V_{rms}$)

Thus the average power can be written as, $P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$

Power absorbed by a resistor can be written as, $P = I_{rms}^2 R = \frac{V_{rms}^2}{R}$

Problem:

Determine the rms value of the current waveform in Fig. 11.14. If the current is passed through a $2\text{-}\Omega$ resistor, find the average power absorbed by the resistor.

Similarly: Practice problems and examples.

Example 11.7

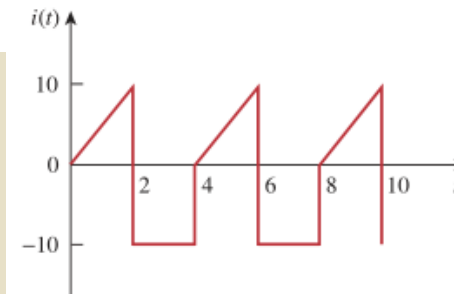


Figure 11.14

Apparent Power and Power Factor

- From previous slide we got, $P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$
- “The apparent power (in VA) is the product of the rms values of voltage and current.”
- Mathematically, $S = V_{rms} I_{rms}$
- It is apparent with compared to dc circuit that power is the multiplication of voltage and current, hence the name apparent power. It is measured in volt-amperes (VA) by which it is different form other powers.
- The factor $\cos(\theta_v - \theta_i)$ is called power factor (pf). It is dimensionless.
- “The power factor is the cosine of the phase difference between voltage and current. I is also the cosine of the angle of the load impedance.”
- The angle $(\theta_v - \theta_i)$ is called power factor angle. (Note that it is always $\theta_v - \theta_i$, not $\theta_i - \theta_v$)
- Again “The ratio of the average power to the apparent power is called power factor”.
- Mathematically, $pf = \frac{P}{S} = \cos(\theta_v - \theta_i)$
- “The power factor angle is equal to the angle of the load impedance”.
- Mathematically, $Z = \frac{V}{I} = \frac{V_m}{I_m} \angle(\theta_v - \theta_i) = \frac{V_{rms}}{I_{rms}} \angle(\theta_v - \theta_i) =$

Problems

Example 11.9

A series-connected load draws a current $i(t) = 4 \cos(100\pi t + 10^\circ)$ A when the applied voltage is $v(t) = 120 \cos(100\pi t - 20^\circ)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

Practice Problem 11.9

Obtain the power factor and the apparent power of a load whose impedance is $\mathbf{Z} = 60 + j40 \Omega$ when the applied voltage is $v(t) = 320 \cos(377t + 10^\circ)$ V.

Similarly, You can see example 11.10 and practice problem 11.10

Complex Power

If, $v = V_m \angle \theta_v$ and $i = I_m \angle \theta_i$

$$\text{Then, } S = \frac{1}{2} VI^* = V_{rms} I_{rms}^* = V_{rms} I_{rms} \angle(\theta_v - \theta_i) = V_{rms} I_{rms} \cos(\theta_v - \theta_i) + j V_{rms} I_{rms} \sin(\theta_v - \theta_i)$$

If load impedance is Z , then $Z = \frac{V}{I} = \frac{V_{rms}}{I_{rms}} \angle(\theta_v - \theta_i)$ & $V_{rms} = Z I_{rms}$

$$\text{Thus, } \mathbf{S} = I_{rms}^2 \mathbf{Z} = \frac{V_{rms}^2}{Z^*} = V_{rms} I_{rms}^* = I_{rms}^2 (R + jX) = (P + jQ)$$

$$\text{Thus, } P = \text{Re}(\mathbf{S}) = I_{rms}^2 R = V_{rms} I_{rms} \cos(\theta_v - \theta_i) \quad \text{and} \quad Q = \text{Im}(\mathbf{S}) = I_{rms}^2 X = V_{rms} I_{rms} \sin(\theta_v - \theta_i)$$

- The real power P is the average power in watts delivered to a load; it is the only useful power. It is the actual power dissipated by the load.
- The reactive power Q is a measure of the energy exchange between the source and the reactive part of the load. The unit of Q is the volt-ampere reactive (VAR)
- The reactive power is being transferred back and forth between the load and the source. It represents a lossless interchange between the load and the source

Complex Power contd...

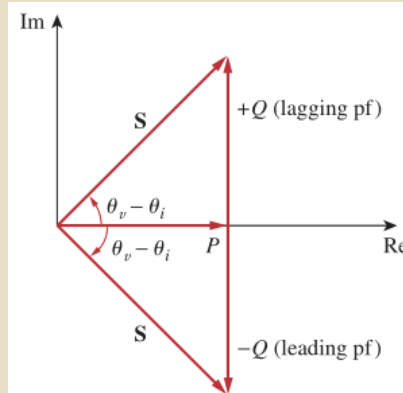
“Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q .

1. $Q=0$ for resistive loads (unity pf).
2. $Q<0$ for capacitive loads (leading pf).
3. $Q>0$ for inductive loads (lagging pf).

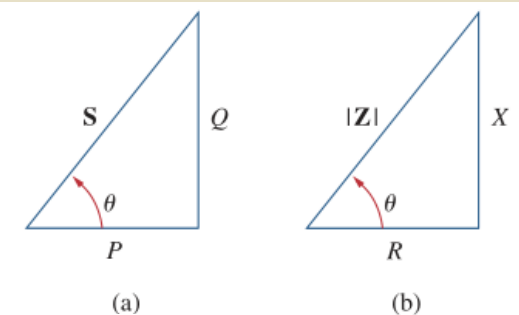
It is a standard practice to represent S , P and Q in the form of a triangle, known as power triangle. It is similar to the impedance triangle. Power triangle contains 4 items. The apparent power, real power, reactive power and power factor angle.

N:B: When ‘ S ’ lies in the first quadrant, we have an inductive load and a lagging pf. When S lies in the fourth quadrant, the load is capacitive and the pf is leading. It is also possible for the complex power to lie in the second or third quadrant. This requires that the load impedance have a negative resistance, which is possible with active circuits.

$$\begin{aligned}\text{Complex Power} = S &= P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^* \\ &= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i \\ \text{Apparent Power} = S &= |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2} \\ \text{Real Power} = P &= \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i) \\ \text{Reactive Power} = Q &= \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i) \\ \text{Power Factor} &= \frac{P}{S} = \cos(\theta_v - \theta_i)\end{aligned}$$



Power triangle.



(a) Power triangle, (b) impedance triangle.

Problems

The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ)$ V and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ)$ A. Find: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Example 11.11

Practice Problem 11.11

For a load, $\mathbf{V}_{\text{rms}} = 110 \angle 85^\circ$ V, $\mathbf{I}_{\text{rms}} = 0.4 \angle 15^\circ$ A. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Example 11.12

A load $\mathbf{Z}$ draws 12 kVA at a power factor of 0.856 lagging from a 120-V rms sinusoidal source. Calculate: (a) the average and reactive powers delivered to the load, (b) the peak current, and (c) the load impedance.

A sinusoidal source supplies 100 kVAR reactive power to load $\mathbf{Z} = 250 \angle -75^\circ \Omega$. Determine: (a) the power factor, (b) the apparent power delivered to the load, and (c) the rms voltage.

Practice Problem 11.12

In both series or parallel connected load,

$$S = S_1 + S_2 + S_3 + \dots + S_N$$

Four things come not back: the spoken word;
the sped arrow; time past; the neglected
opportunity.

Omar Ibn Al Hanif